

DOOR COUNTY CHERRYLAND AP, WI, USA

WMO: 726458

Lat: 44.850N

Lon: 87.417W

Elev: 221

StdP: 98.70

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.0	-27.2	0.3	-20.1	-23.8	0.5	-18.1	11.7	-7.4	10.9	-6.2	4.5	260	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.0	22.0	27.6	21.0	26.3	20.0	23.5	27.7	22.3	26.3	21.2	24.8	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.3	26.7	21.0	16.1	25.2	19.8	14.9	23.7	70.8	27.8	66.5	26.4	62.5	25.2	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.6	DB	-23.8	32.4	3.7	1.9	-26.4	33.8	-28.6	34.9	-30.7	36.0	-33.3	37.4	(4)
(5)			WB	-23.9	25.5	2.9	1.1	-26.0	26.3	-27.7	26.9	-29.4	27.6	-31.5	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-7.2	-6.0	-1.0	5.4	11.9	17.5	20.5	19.8	16.2	9.7	2.7	-3.4
	DBStd	10.81	6.31	6.17	5.61	4.44	4.30	3.82	3.13	2.93	4.00	4.44	4.83	5.53
	HDD10.0	2201	532	447	342	148	28	1	0	0	2	62	222	417
	HDD18.3	4253	791	680	599	387	205	59	14	18	87	271	468	675
	CDD10.0	1201	0	0	2	11	87	227	327	304	189	51	4	0
	CDD18.3	212	0	0	0	0	6	35	83	63	23	2	0	0
	CDH23.3	1209	0	0	0	1	46	224	538	297	98	6	0	0
	CDH26.7	256	0	0	0	0	7	46	135	51	17	1	0	0
Wind	WSAvg	4.3	4.6	4.6	4.5	4.7	4.2	3.7	3.5	3.4	4.0	4.6	4.9	4.7
Precipitation	PrecAvg	793	44	30	46	70	87	99	88	80	79	72	52	44
	PrecMax	957	102	53	103	129	155	226	167	149	169	164	138	96
	PrecMin	626	9	11	10	14	38	26	26	11	31	16	8	6
	PrecStd	79	26	12	24	29	32	58	40	30	34	42	31	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	8.7	16.0	21.3	27.4	30.2	32.4	30.9	29.0	23.7	17.3	11.2
		MCWB	4.6	6.0	11.2	14.0	19.6	22.9	24.5	22.1	21.7	18.3	12.0	9.8
	2%	DB	4.0	5.9	11.4	17.4	24.1	27.8	29.8	27.8	26.2	21.0	13.1	7.3
		MCWB	2.3	3.4	7.8	10.9	17.1	20.5	22.5	21.5	19.8	16.4	9.8	5.3
	5%	DB	2.5	3.5	8.5	14.2	22.1	26.2	27.8	26.4	23.9	17.9	11.2	5.1
		MCWB	1.3	1.7	5.6	8.6	16.0	19.5	21.4	20.3	18.8	14.3	8.6	3.4
	10%	DB	1.3	2.2	6.3	12.3	19.0	23.9	26.6	24.9	22.4	16.3	9.0	3.1
		MCWB	0.2	0.8	3.4	7.5	14.0	18.4	20.4	19.5	18.0	13.3	7.0	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	6.6	11.9	14.9	20.7	24.1	25.5	24.4	22.9	19.7	12.9	9.8
		MCDB	6.7	8.4	15.0	19.5	25.9	28.4	31.0	28.4	27.7	22.5	15.9	11.0
	2%	WB	2.9	3.6	8.1	12.0	18.5	21.9	23.8	22.9	21.0	17.0	10.7	5.7
		MCDB	3.9	5.3	10.4	15.9	23.0	26.3	28.1	26.5	24.4	19.6	13.0	7.0
	5%	WB	1.4	2.0	5.6	9.9	16.6	20.4	22.6	21.7	19.9	15.1	8.7	3.7
		MCDB	2.4	3.3	8.2	13.4	20.6	24.7	26.8	25.1	23.0	17.7	10.4	5.0
	10%	WB	0.2	0.6	3.7	7.9	14.8	19.1	21.4	20.6	18.8	13.5	7.4	2.2
		MCDB	1.2	1.9	6.1	11.8	18.9	23.1	25.2	23.6	21.6	16.1	8.9	3.4
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	7.6	8.3	9.3	9.0	9.3	8.2	8.3	6.9	5.8	5.7
		MCDBR	7.0	6.8	9.8	12.3	12.8	10.9	10.6	9.5	10.0	9.2	8.5	7.0
		MCWBR	6.3	5.6	7.2	8.1	8.1	6.5	5.8	5.3	5.8	6.5	6.6	6.3
	5% WB	MCDBR	6.6	6.2	9.2	11.8	12.0	10.0	9.8	8.3	8.7	8.7	7.7	6.5
		MCWBR	6.1	5.4	7.3	8.5	8.5	6.8	6.1	5.3	5.7	6.6	6.6	6.2
Clear-Sky Solar Irradiance	taub	0.270	0.293	0.325	0.367	0.395	0.403	0.393	0.393	0.370	0.332	0.303	0.271	
	taud	2.465	2.384	2.392	2.335	2.320	2.337	2.378	2.372	2.429	2.520	2.527	2.512	
	Ebn at Noon	862	897	910	893	875	867	871	858	851	844	815	827	
	Edh at Noon	69	90	103	119	125	123	117	113	99	79	65	60	
All-Sky Solar Radiation	RadAvg	1.41	2.39	3.60	4.53	5.52	5.98	6.13	5.22	4.06	2.50	1.50	1.09	
	RadStd	0.16	0.25	0.33	0.43	0.45	0.45	0.25	0.34	0.31	0.25	0.15	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page